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(54) **PHOTOACOUSTIC SENSORS AND MEMS DEVICES**

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B81B 3/00 (2006.01)

(52) **U.S. Cl.**

CPC **G01N 29/02** (2013.01); **B81B 3/0018** (2013.01); **B81B 2201/0292** (2013.01); **G01N 2291/021** (2013.01)

(58) **Field of Classification Search**

CPC G01N 29/02; G01N 2291/021; G01N 2021/0112; G01N 2021/1704; G01N 21/1702; B81B 3/0018; B81B 2201/0292
See application file for complete search history.

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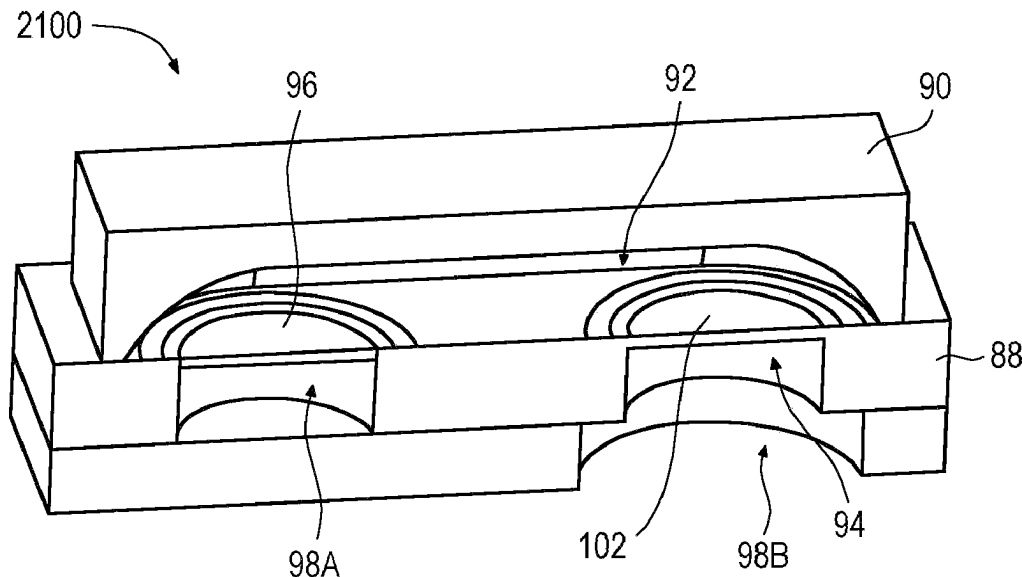
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(57) **ABSTRACT**

A photoacoustic sensor includes a first MEMS device and a second MEMS device. The first MEMS device includes a first MEMS component including an optical emitter, and a first optically transparent cover wafer-bonded to the first MEMS component, wherein the first MEMS component and the first optically transparent cover form a first closed cavity. The second MEMS device includes a second MEMS component including a pressure detector, and a second optically transparent cover wafer-bonded to the second MEMS component, wherein the second MEMS component and the second optically transparent cover form a second closed cavity.

20 Claims, 9 Drawing Sheets



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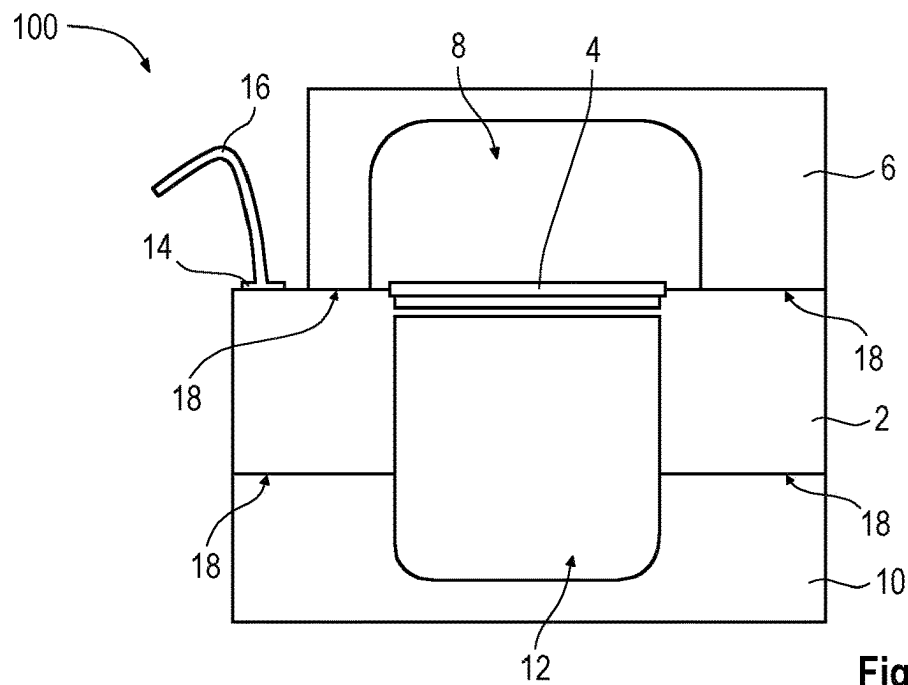


Fig. 1

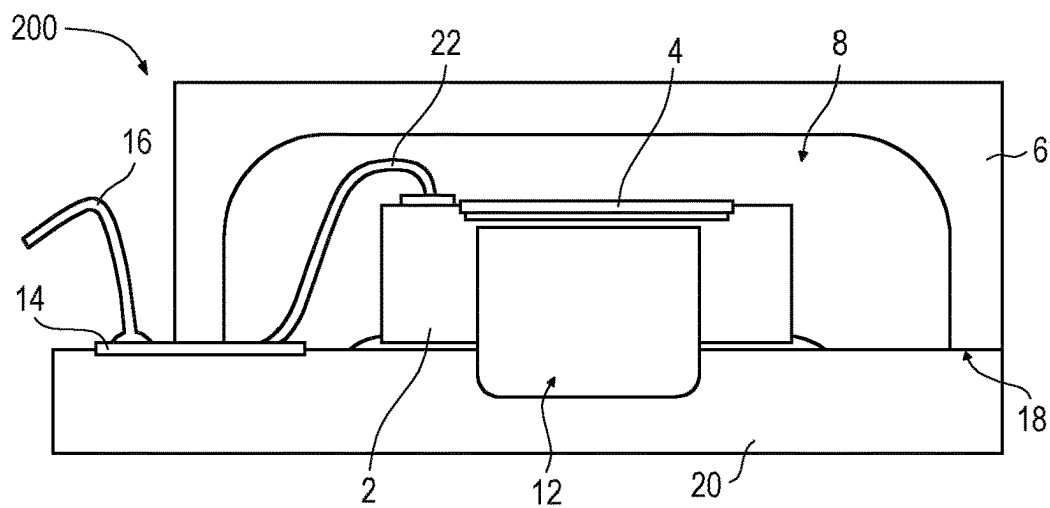


Fig. 2

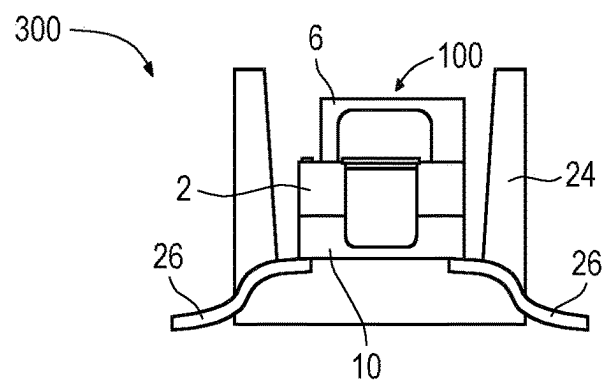


Fig. 3

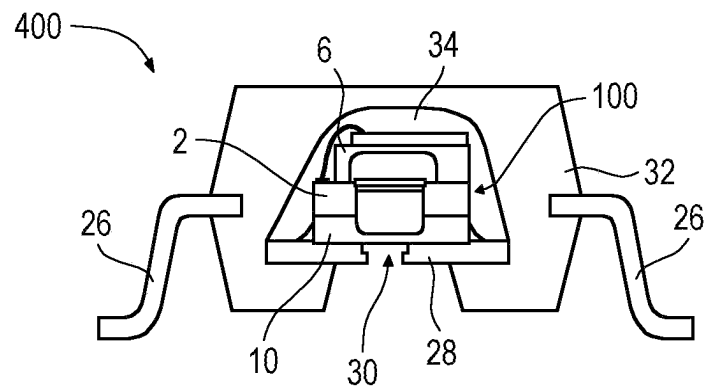


Fig. 4

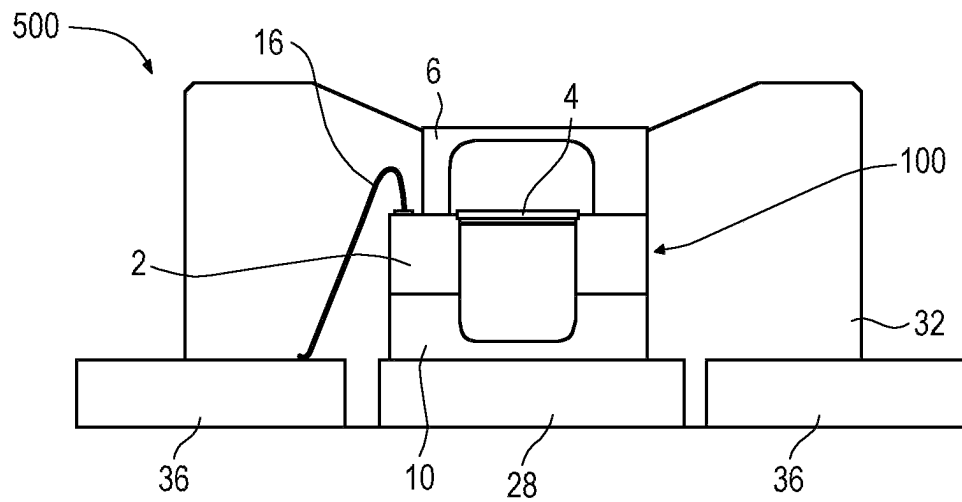


Fig. 5

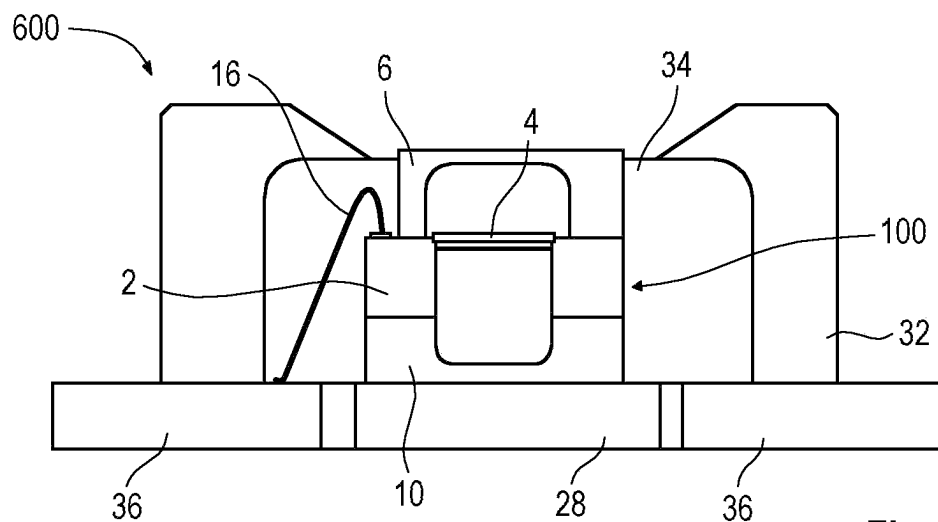


Fig. 6

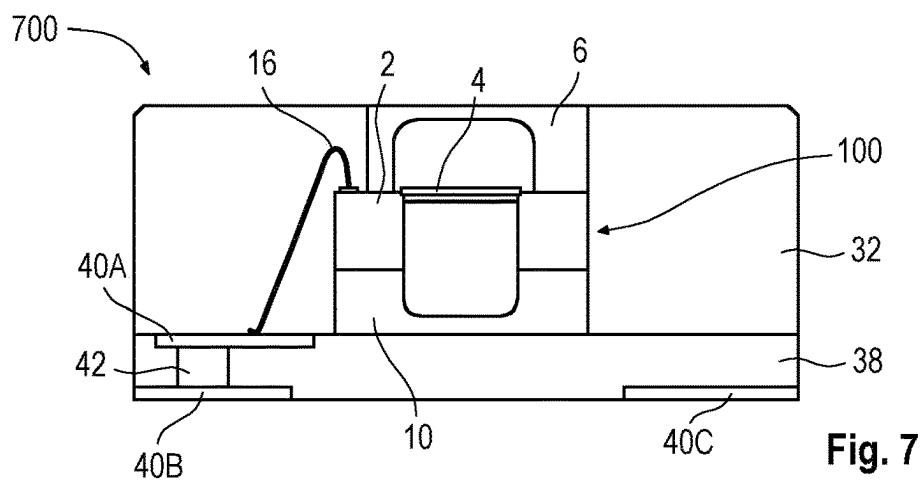


Fig. 7

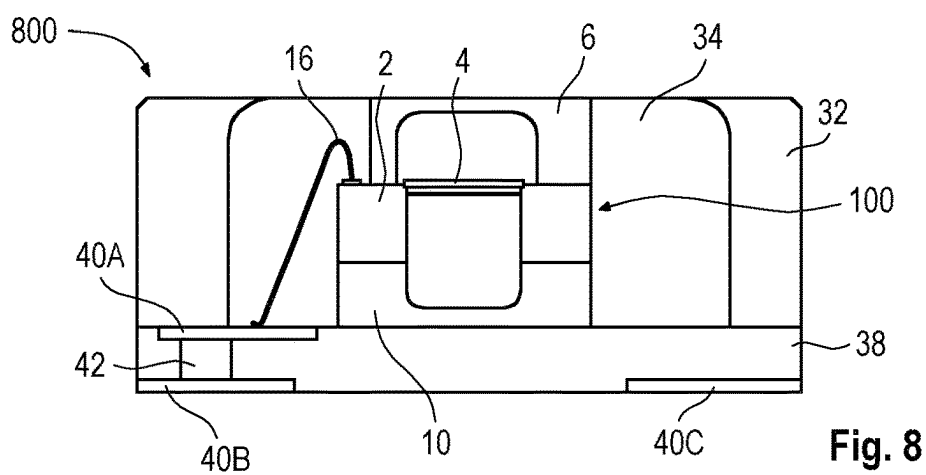


Fig. 8

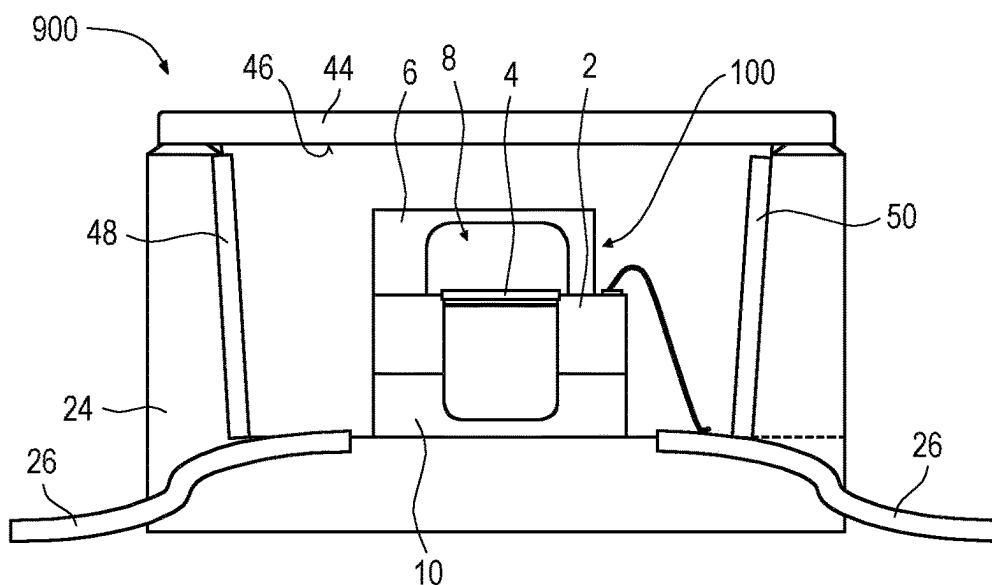


Fig. 9

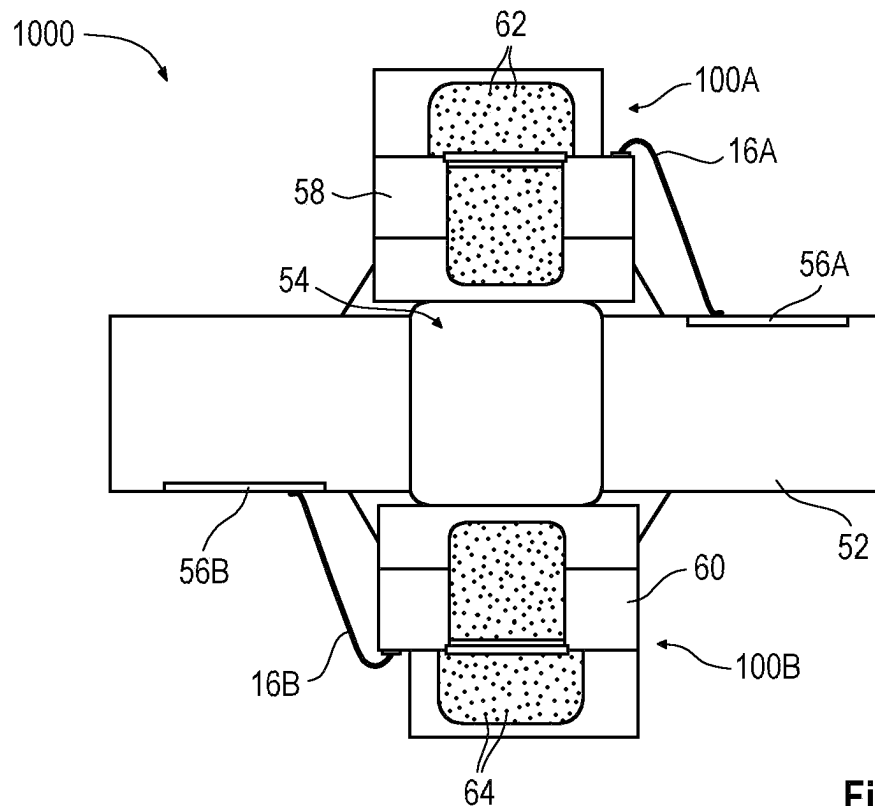


Fig. 10

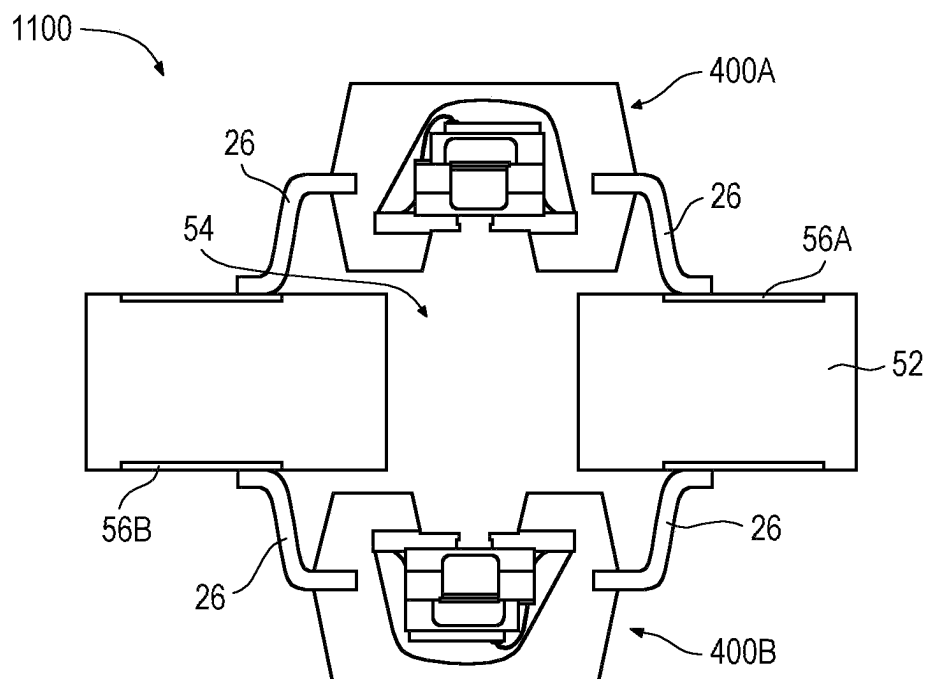


Fig. 11

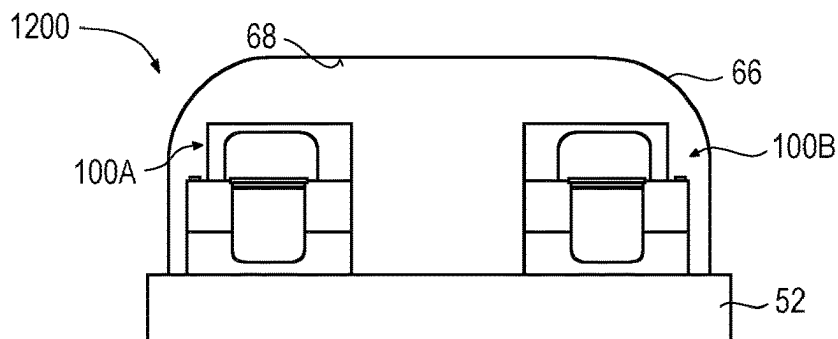


Fig. 12

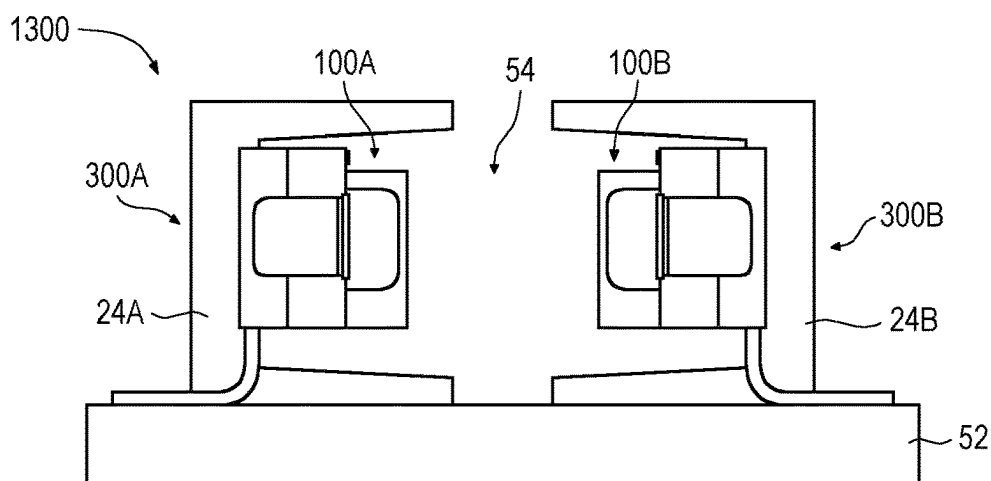


Fig. 13

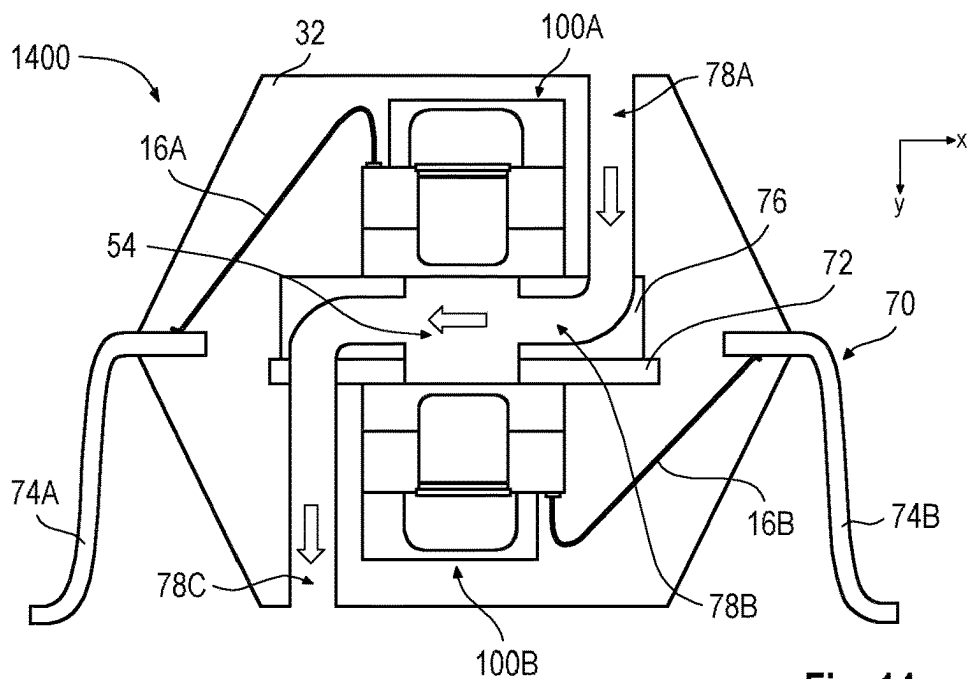


Fig. 14

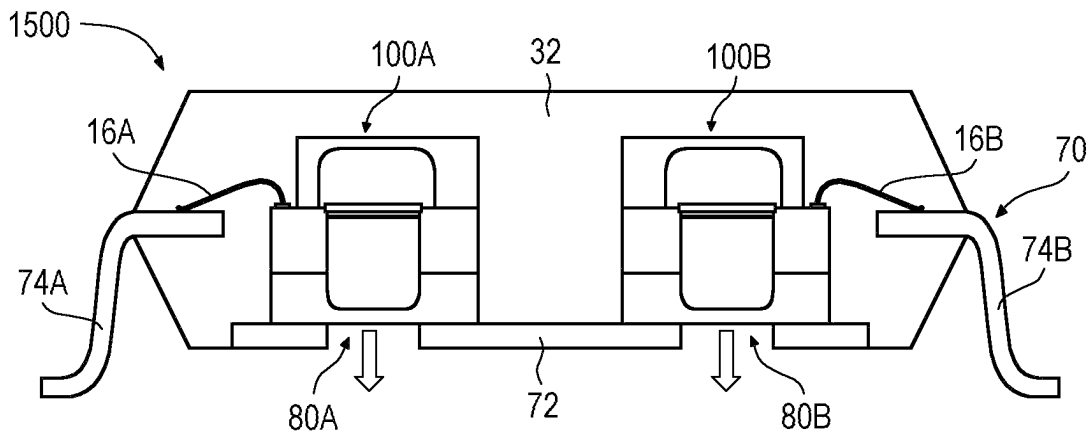


Fig. 15

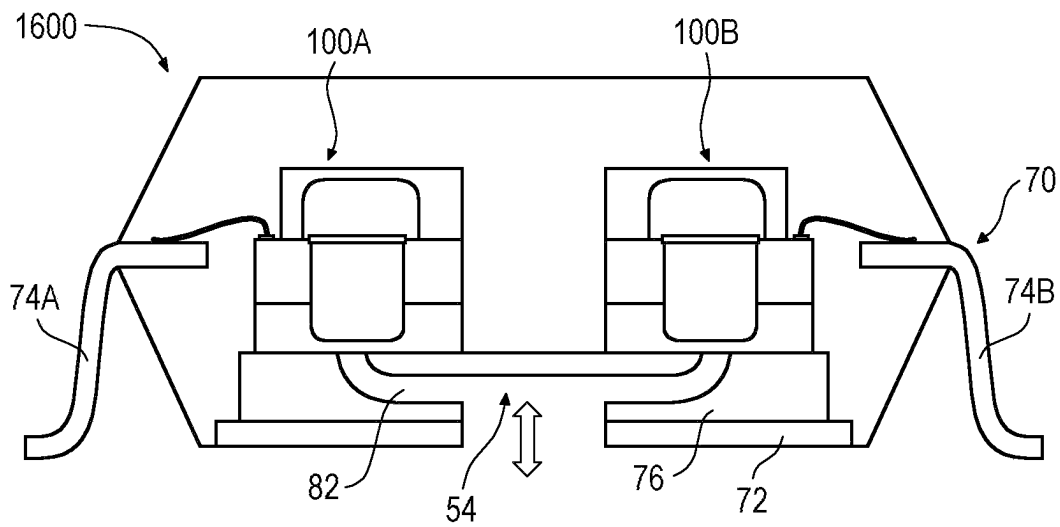


Fig. 16

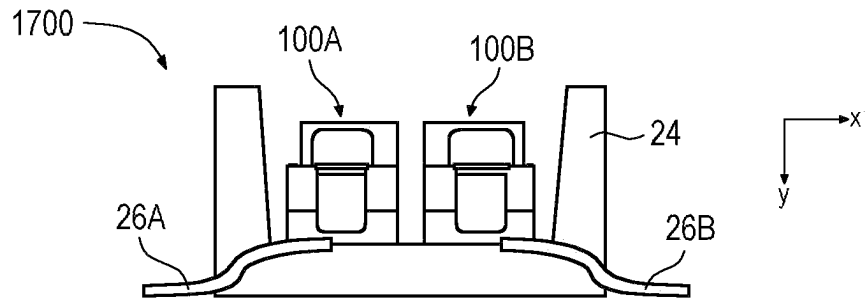


Fig. 17

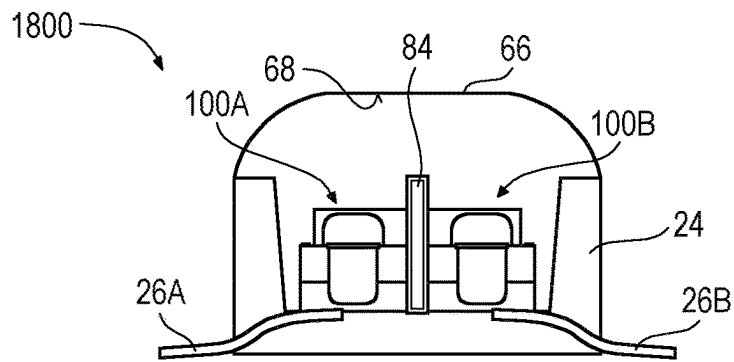


Fig. 18

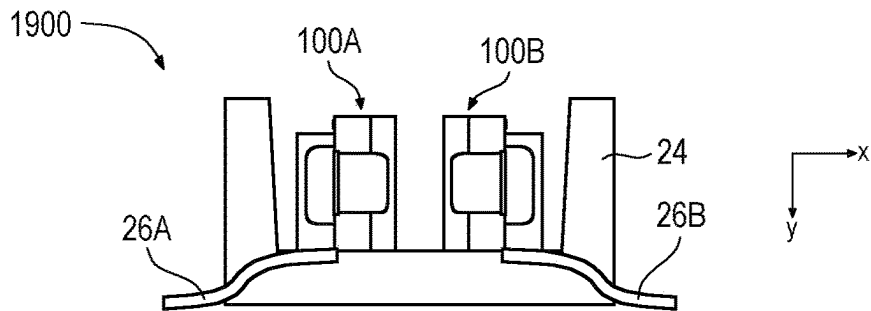


Fig. 19

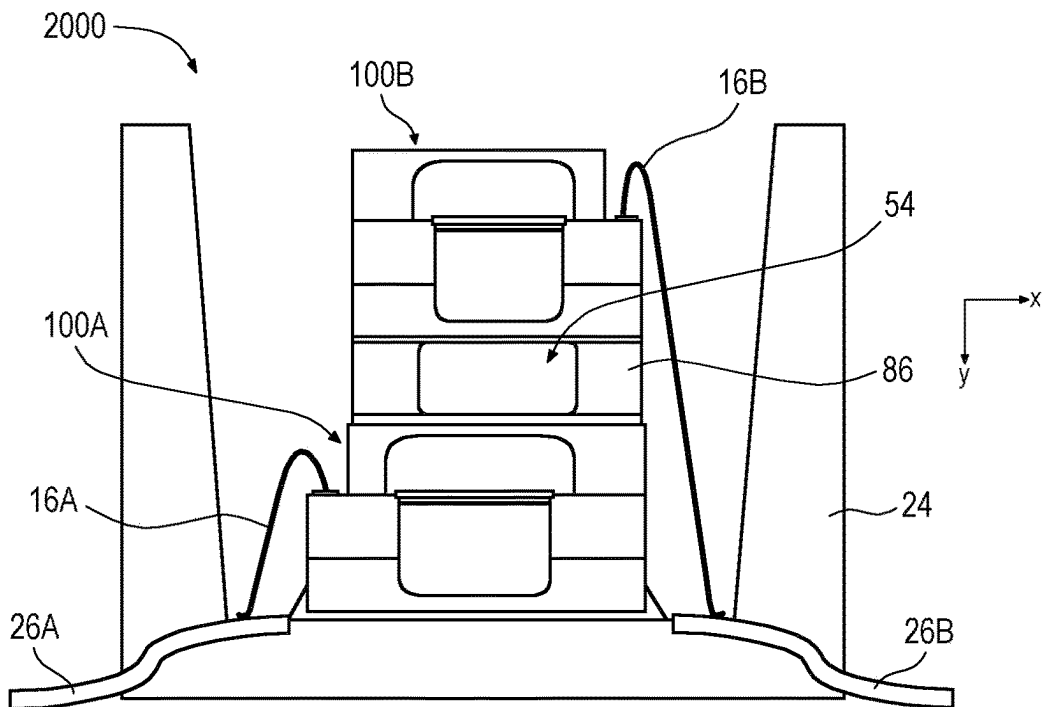


Fig. 20

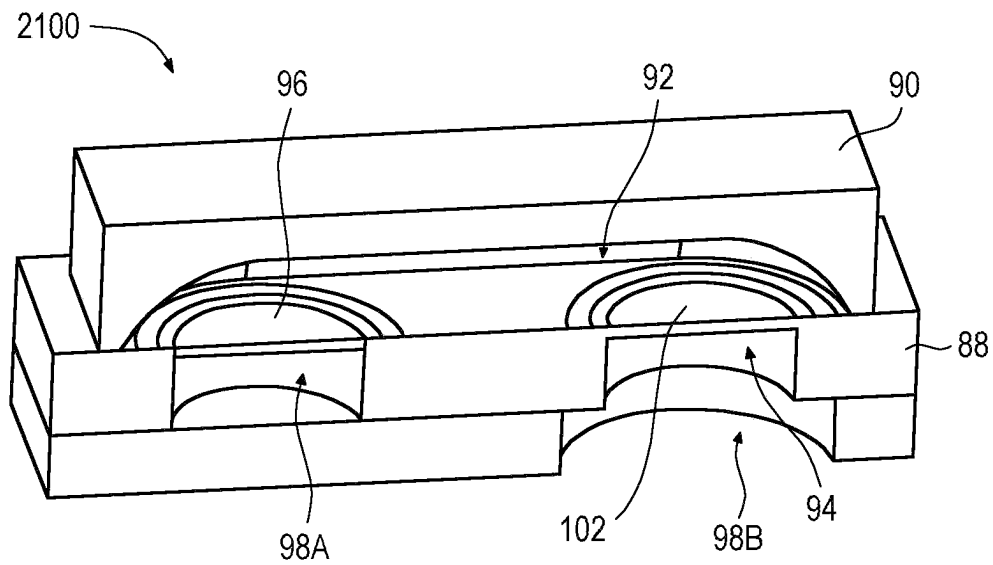


Fig. 21

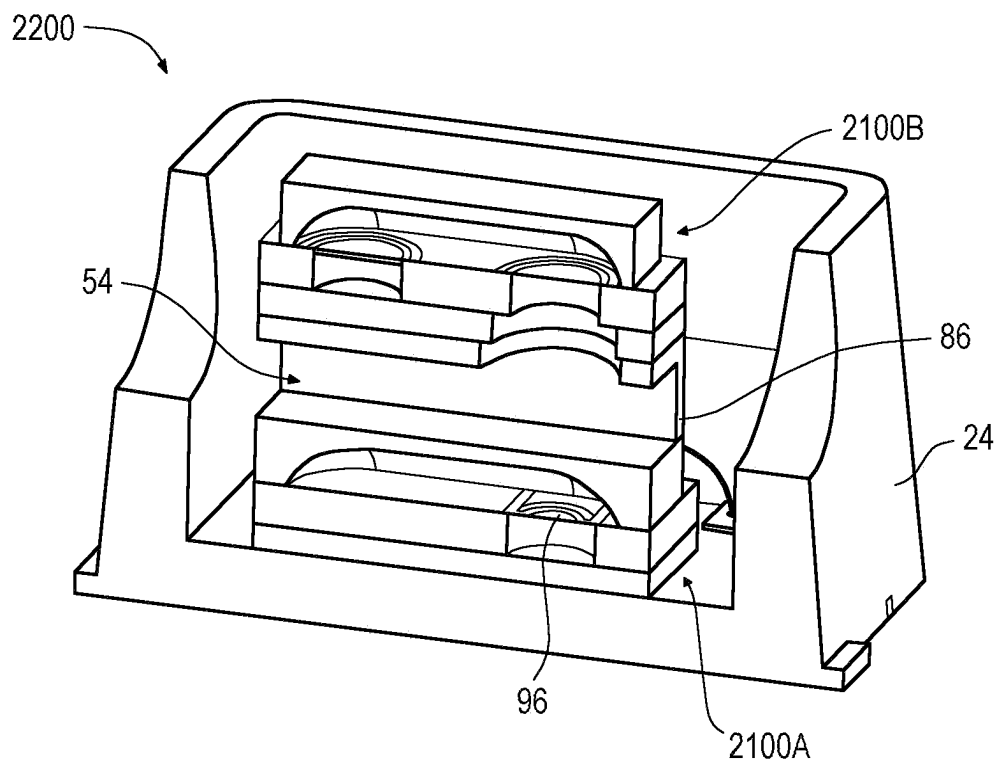


Fig. 22

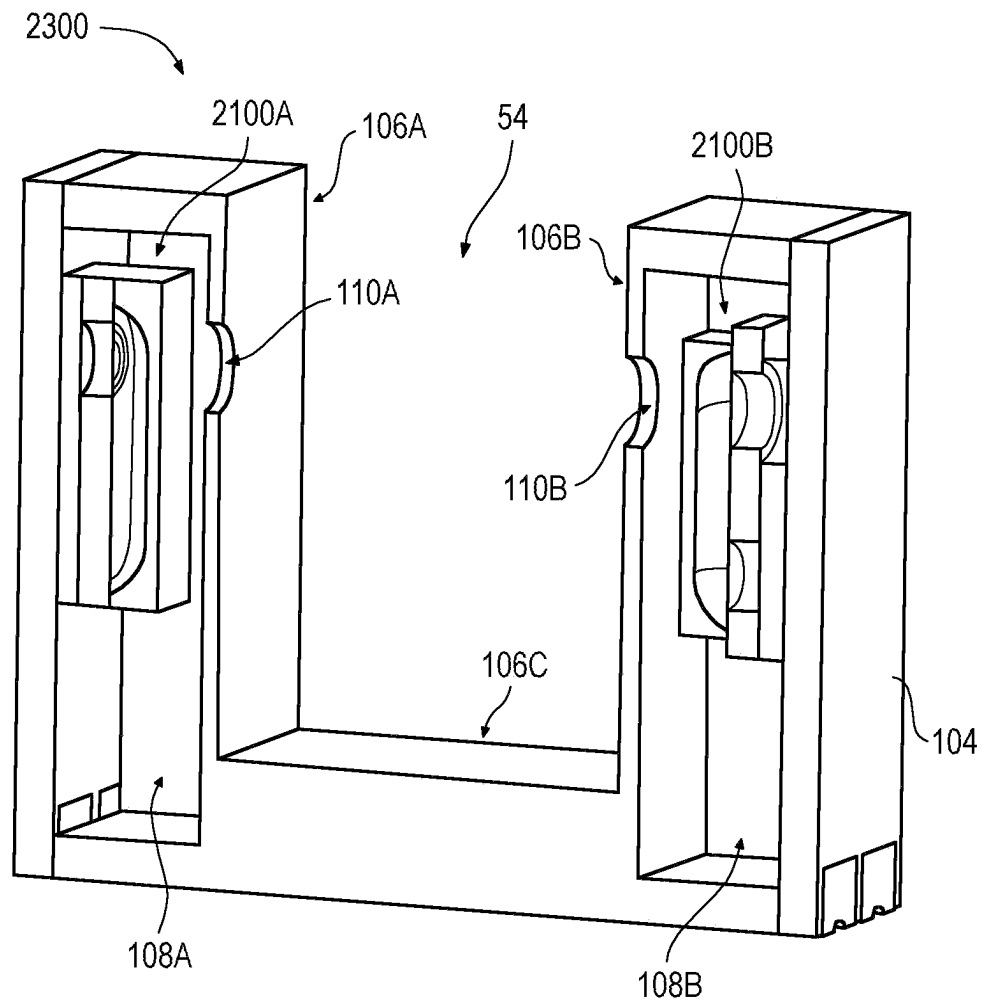


Fig. 23

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PHOTOACOUSTIC SENSORS AND MEMS DEVICES

RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 17/116,029, filed Dec. 9, 2020, which claims priority to Germany Application No. 102019134279.1, filed Dec. 13, 2019, which are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates to photoacoustic sensors and MEMS devices.

BACKGROUND

Photoacoustic sensors can detect specific gas species in the ambient air, for example. In particular, harmful or hazardous components in the ambient air can be detected in this case. The correct functioning of such sensors can thus be of extremely high importance in many applications, particularly if the sensors are intended to guarantee the safety of work personnel. Photoacoustic sensors can be constructed from a plurality of components, in particular MEMS devices. Manufacturers of photoacoustic sensors and MEMS devices constantly endeavor to improve their products. In particular, it may be desirable in this case to provide cost-effective photoacoustic sensors having an improved design. Furthermore, it may be desirable to provide improved methods for producing such apparatuses.

SUMMARY

Various aspects relate to a photoacoustic sensor. The photoacoustic sensor includes a first MEMS device and a second MEMS device. The first MEMS device includes a first MEMS component including an optical emitter, and a first optically transparent cover wafer-bonded to the first MEMS component, wherein the first MEMS component and the first optically transparent cover form a first closed cavity. The second MEMS device includes a second MEMS component including a pressure detector, and a second optically transparent cover wafer-bonded to the second MEMS component, wherein the second MEMS component and the second optically transparent cover form a second closed cavity.

Various aspects relate to a MEMS device. The MEMS device includes a MEMS component and a cover secured to the MEMS component, wherein the MEMS component and the cover form a closed cavity. The MEMS device furthermore includes an optical opening, which provides an optical access to the cavity and to an optical path extending within the cavity. A movable part of the MEMS component is arranged outside the course of the optical path.

BRIEF DESCRIPTION OF THE DRAWINGS

Photoacoustic sensors and MEMS devices in accordance with the disclosure are explained in greater detail below with reference to drawings. The elements shown in the drawings are not necessarily rendered in a manner true to scale relative to one another. Identical reference signs may designate identical components.

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FIG. 1 schematically illustrates a cross-sectional side view of a MEMS device **100** in accordance with the disclosure.

FIG. 2 schematically illustrates a cross-sectional side view of a MEMS device **200** in accordance with the disclosure.

FIG. 3 schematically illustrates a cross-sectional side view of a packaged MEMS device **300** in accordance with the disclosure.

FIG. 4 schematically illustrates a cross-sectional side view of a packaged MEMS device **400** in accordance with the disclosure.

FIG. 5 schematically illustrates a cross-sectional side view of a packaged MEMS device **500** in accordance with the disclosure.

FIG. 6 schematically illustrates a cross-sectional side view of a packaged MEMS device **600** in accordance with the disclosure.

FIG. 7 schematically illustrates a cross-sectional side view of a packaged MEMS device **700** in accordance with the disclosure.

FIG. 8 schematically illustrates a cross-sectional side view of a packaged MEMS device **800** in accordance with the disclosure.

FIG. 9 schematically illustrates a cross-sectional side view of a packaged MEMS device **900** in accordance with the disclosure.

FIG. 10 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1000** in accordance with the disclosure.

FIG. 11 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1100** in accordance with the disclosure.

FIG. 12 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1200** in accordance with the disclosure.

FIG. 13 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1300** in accordance with the disclosure.

FIG. 14 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1400** in accordance with the disclosure.

FIG. 15 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1500** in accordance with the disclosure.

FIG. 16 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1600** in accordance with the disclosure.

FIG. 17 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1700** in accordance with the disclosure.

FIG. 18 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1800** in accordance with the disclosure.

FIG. 19 schematically illustrates a cross-sectional side view of a photoacoustic sensor **1900** in accordance with the disclosure.

FIG. 20 schematically illustrates a cross-sectional side view of a photoacoustic sensor **2000** in accordance with the disclosure.

FIG. 21 schematically illustrates a perspective cross-sectional view of a MEMS device **2100** in accordance with the disclosure.

FIG. 22 schematically illustrates a perspective cross-sectional view of a photoacoustic sensor **2200** in accordance with the disclosure.

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FIG. 23 schematically illustrates a perspective cross-sectional view of a photoacoustic sensor 2300 in accordance with the disclosure.

DETAILED DESCRIPTION

The figures described below show photoacoustic sensors and MEMS devices in accordance with the disclosure. In this case, the described apparatuses may be illustrated in a general way in order to describe aspects of the disclosure qualitatively. The apparatuses described may have further aspects that may not be illustrated in the respective figure for the sake of simplicity. However, the respective example may be extended by aspects described in association with other examples in accordance with the disclosure. Consequently, explanations concerning a specific figure may equally apply to examples of other figures.

FIGS. 1 and 2 show unpackaged MEMS devices 100 and 200. The MEMS devices 100 and 200 described can each form a photoacoustic emitter unit or a photoacoustic detector unit. A photoacoustic sensor, for example, can thus be constructed from the MEMS devices 100 and 200.

The MEMS device 100 in FIG. 1 can comprise a MEMS component 2 having one or more movable structures (MEMS structures 4). The MEMS component 2 can be a MEMS chip, which can be fabricated from a semiconductor material such as silicon, for example. In one example, the MEMS component 2 can comprise a pressure detector. The pressure detectors specified in this description can be, for example, microphones or any other type of pressure sensors or pressure-sensitive sensors. In this case, the MEMS device 100 can form for example a photoacoustic detector unit of a photoacoustic sensor. In a further example, the MEMS component 2 can comprise an emitter of optical radiation, wherein the movable structure 4 can form a heating membrane, for example. In this case, the MEMS device 100 can form for example a photoacoustic emitter unit of a photoacoustic sensor.

The MEMS device 100 can comprise an optically transparent cover 6, which can be secured to the top side of the MEMS component 2. In this case, the MEMS component 2 and the optically transparent cover 6 can be wafer-bonded, in particular, that is to say that a connection of the two components may have been carried out at the wafer level. The optically transparent cover 6 can be fabricated for example from a glass material, a semiconductor material, or a ceramic material. The MEMS component 2 and the optically transparent cover 6 can form a first closed cavity 8. Depending on the application of the MEMS device 100, a protective gas or a reference gas can be enclosed in the cavity 8. The protective gases specified in this description can be for example nitrogen or a noble gas, such as e.g. argon, xenon, krypton. The reference gases specified in this description can be for example carbon dioxide, nitrogen oxide, methane, ammonia.

The MEMS device 100 can comprise a cover 10, which can be secured to the underside of the MEMS component 2. In this case, the MEMS component 2 and the cover 10 can be wafer-bonded, in particular, that is to say that a connection of the two components may have been carried out at the wafer level. The cover 10 can be optically transparent. By way of example, the cover 10 can be fabricated from a glass material, a semiconductor material, or a ceramic material. The MEMS component 2 and the cover 10 can form a further closed cavity 12. Depending on the application of the MEMS device 100, a protective gas or a reference gas can be enclosed in the cavity 12.

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The MEMS component 2 can comprise an electrical connection 14, which can be electrically contacted by way of an electrical connection element 16 such as a bond wire, for example. The MEMS component 2 can be electrically connected to external components (not illustrated) by way of the electrical connection 14 and the electrical connection element 16. If the MEMS device 100 forms a photoacoustic detector unit, for example, signals detected by the movable structure 4 can be transmitted to an external component in this way. If the MEMS device 100 forms a photoacoustic emitter unit, for example, the optical radiation emitted by the movable structure 4 can be controlled by an external component in this way.

The covers 6 and 10 can each be wafer-bonded to the MEMS component 2. The connections between the respective components can thus be present in the form of so-called wafer bonds 18. Different wafer bonding techniques can be used here depending on the material of the respective cover 6 or 10 and the MEMS component 2. One example can involve wafer bonding without the use of an intermediate layer. This can involve in particular direct bonding or anodic bonding. A further example can involve wafer bonding using an intermediate layer. This can involve in particular glass frit bonding, soldering, eutectic bonding, thermocompression bonding, or adhesive bonding.

During the production of the MEMS device 100, the parts thereof can be connected at the wafer level. That is to say at least one of the components 2, 6 and 10 can initially be present in the form of a wafer during the production of the MEMS device 100. By way of example, the cover 6 can initially be part of a glass wafer that can comprise any desired number of further covers. The wafers can be connected to one another using one of the wafer bonding techniques mentioned above. The bonded wafers can subsequently be singulated into a plurality of MEMS devices 100. Production of the MEMS device 100 at the wafer level using the wafer bonding techniques described above may be more cost-effective in comparison with other production methods.

The MEMS device 200 in FIG. 2 can be at least partly similar to the MEMS device 100 in FIG. 1. Explanations concerning FIG. 1 can thus also be applicable to the MEMS device 200. The MEMS device 200 can comprise a carrier 20, on which the MEMS component 2 can be arranged. By way of example, the carrier 20 can be a semiconductor substrate, in particular a silicon substrate. The MEMS component 2 can be secured to the carrier 20 for example by an adhesive or a DDAF (Dicing Die Attach Film). A cavity 12 can be formed between the underside of the movable structure 4 and the top side of the carrier 20.

Furthermore, the MEMS device 200 can comprise an optically transparent cover 6, which can be secured to the top side of the carrier 20. In this case, the optically transparent cover 6 and the carrier 20 can be wafer-bonded, in particular, such that a connection between these components can be present in the form of a wafer bond 18. The optically transparent cover 6 and the carrier 20 can form a cavity 8, in which the MEMS component 2 can be arranged. In the example in FIG. 2, the carrier 20 can be embodied as substantially planar. In further examples, the carrier 20, in particular that part of the carrier 20 which is arranged below the cover 6, can have a depression. The depression can be formed in the shape of a trough, for example. The MEMS component 2 can be arranged in the depression, such that the movable part 4 can be at a deeper level and/or the height of the cavity 8 can be increased in comparison with FIG. 2. The MEMS component 2 can be electrically connected, by way

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of an electrical connection element **22**, to an electrical connection **14** arranged on the carrier **20** and to a further electrical connection element **16**. Being electrically connected in this way enables the MEMS component **2** to transmit or receive electrical signals to or from external components.

FIGS. **3** to **9** show packaged MEMS devices **300** to **900**. The MEMS devices **300** to **900** described can each form a photoacoustic emitter unit or a photoacoustic detector unit. A photoacoustic sensor, for example, can thus be constructed from the MEMS devices **300** to **900**. FIGS. **3** to **9** show in particular possibilities for packaging the MEMS devices **100** and **200** from FIGS. **1** and **2**. In this case, the examples shown in FIGS. **3** to **9** are based on the MEMS device **100** from FIG. **1**. Further examples with the same construction can be based on the MEMS device **200** from FIG. **2**.

The MEMS device **300** in FIG. **3** can comprise a package in the form of a shell or trough **24**. In one example, the shell **24** can be fabricated from a mold compound. The mold compounds specified in this description can comprise at least one from an epoxy, a filled epoxy, a glass-fiber-filled epoxy, an imide, a thermoplastic, a thermosetting polymer, a polymer mixture. An unpackaged MEMS device **100** can be arranged on a base surface of the shell **24**. The MEMS device **100** can optionally additionally be embedded into an encapsulation material such as, for example, a glob top material (not illustrated). The packaged MEMS device **300** can furthermore comprise one or more connecting conductors **26**. The connecting conductors **26** can extend through the walls of the shell **24** and provide an electrical connection between the MEMS component **2** and components (not illustrated) arranged outside the shell **24**.

The packaged MEMS device **400** in FIG. **4** can comprise a carrier **28**, which can be for example a die pad of a leadframe. An unpackaged MEMS device **100** can be arranged on the carrier **28**. An opening **30** can be formed in the carrier **28**, through which opening the MEMS device **100** can emit and/or receive optical radiation. The MEMS device **100** can be at least partly encapsulated in an encapsulation material **32**, which can be fabricated from a mold compound, in particular. Within the encapsulation material **32**, the unpackaged MEMS device **100** can be optionally additionally embedded in a further encapsulation material **34** such as, for example, a glob top material.

The packaged MEMS device **400** can comprise one or more connecting conductors **26**, which can project at least partly from the encapsulation material **32**. The connecting conductors **26** can be for example leads or pins of a leadframe. The MEMS component **2** can be electrically contacted from outside the encapsulation material **32** by way of the connecting conductors **26**. An electrical connection between the MEMS component **2** and the connecting conductors **26** is not illustrated in FIG. **4** for the sake of simplicity.

The packaged MEMS device **500** in FIG. **5** can comprise a carrier **28** and electrical connections **36**, which can be arranged in an identical plane. In this case, in particular the top sides and undersides of the carrier **28** and of the connections **36** can be arranged in each case in a coplanar fashion. The carrier **28** and the electrical connections **36** can be for example parts of an identical leadframe. A MEMS device **100** can be arranged on the carrier **28**, which MEMS device can be at least partly encapsulated in an encapsulation material **32**. The encapsulation material **32** can be fabricated from a mold compound, in particular. In the example in FIG. **5**, the encapsulation material **32** can cover the side surfaces of the MEMS device **100**. The top side of

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the MEMS device **100** can remain free of the encapsulation material **32**, such that optical radiation can be emitted and/or received by the MEMS device **100**. The encapsulation material **32** can cover the carrier **28** and the electrical connections **36** and can also be arranged in the interspaces of these components. An underside of the encapsulation material **32** and undersides of the carrier **28** and of the electrical connections **36** can be arranged in one plane, i.e. can be embodied in a coplanar fashion.

The packaged MEMS device **600** in FIG. **6** can be at least partly similar to the MEMS device **500** in FIG. **5**. In contrast to FIG. **5**, the MEMS device **600** in FIG. **6** can comprise a further encapsulation material **34**, which can be at least partly covered by the encapsulation material **32**. The further encapsulation material can be a glob top material, for example. The MEMS device **100** can be embedded in the further encapsulation material **34**, wherein a top side of the MEMS device **100** can remain free of the further encapsulation material **34** in order to enable optical radiation to be exchanged between the MEMS device **100** and the surroundings.

The packaged MEMS device **700** in FIG. **7** can comprise a substrate **38**. The substrate **38** can have one or more electrical connections **40A** to **40C** on its top side and underside. The electrical connections **40A** to **40C** can be electrically connected to one another by way of a wiring structure **42** arranged within the substrate **38**. As a result, electrical signals can be redistributed between electrical connections on an identical side of the substrate **38** or on opposite sides of the substrate **38**. The MEMS component **2** can be electrically connected to the electrical connection **40A** on the top side of the substrate **38** by way of an electrical connection element **16**. The electrical connection **40A** can be electrically connected to the electrical connection **40B** on the underside of the substrate **38** by way of the wiring structure **42**. The MEMS component **100** can thus be contacted from outside the packaged MEMS device **700** by way of the electrical connection **40B**. The MEMS component **100** can be at least partly encapsulated in an encapsulation material **32**, which can be fabricated from a mold compound, for example. In this case, in particular the top side of the MEMS device **100** can remain free of the encapsulation material **32**.

The packaged MEMS device **800** in FIG. **8** can be at least partly similar to the MEMS device **700** in FIG. **7**. In contrast to FIG. **7**, the MEMS device **800** in FIG. **8** can comprise a further encapsulation material **34**, which can be covered by the encapsulation material **32**. The further encapsulation material **34** can be a glob top material, for example. The top side of the MEMS device **100** can remain free of both encapsulation materials **32** and **34**.

In a manner similar to FIG. **3**, the MEMS device **900** in FIG. **9** can comprise a package in the form of a shell **24**, a MEMS device **100** arranged therein and connecting conductors **26** extending through the shell **24**. An opening on the top side of the shell **24** can be covered by a cover **44** having an optically reflected inner surface **46**. The shell **24** and the cover **44** can form a closed cavity. An optically reflective structure **48** can be arranged on the left sidewall of the shell **24**, which structure can be realized for example by an optically reflective coating or an optically reflective lamina. An optically transparent window **50** can be formed in the right sidewall of the shell **24**.

In one example, the MEMS device **900** can be operated as a photoacoustic emitter unit. Optical radiation generated by the movable structure **4** can be reflected from the inner surface **46** of the cover **44** and/or the structure **48** in such a

way that the optical radiation can emerge from the interior of the MEMS device **900** through the window **50**. In a further example, the MEMS device **900** can be operated as a photoacoustic detector unit. Optical radiation emitted by an emitter, for example, can enter the interior of the MEMS device **900** through the window **50**. The optical radiation can be reflected from the inner surface **46** of the cover **44** and/or the structure **48** in such a way that the optical radiation can enter the cavity **8** of the MEMS device **100**. Detailed paths of the optical radiation are not illustrated in FIG. **9** for the sake of simplicity.

FIGS. **10** to **20** show photoacoustic sensors **1000** to **2000**, each of which can comprise a photoacoustic emitter unit and a photoacoustic detector unit. The emitter unit and detector unit can be realized here in each case by one of the MEMS devices described in association with previous figures.

The photoacoustic sensor **1000** in FIG. **10** can comprise a substrate **52** having an opening **54**. The substrate **52** can be for example a printed circuit board substrate or a ceramic substrate. The substrate **52** can have one or more electrical connections **56A** and **56B** on its top side and underside. A wiring structure (not illustrated) can be formed within the substrate **52**, and can electrically connect the electrical connections **56A** and **56B** of the substrate **52** to one another in a suitable manner.

An (unpackaged) first MEMS device **100A** can be arranged on the top side of the substrate **52** and over the opening **54**. The first MEMS device **100A** can be a photoacoustic emitter unit, which can comprise an emitter **58** in the form of a MEMS component. The cavities of the first MEMS device **100A** can be filled by a protective gas **62**. The first MEMS device **100A** can be electrically connected to a first electrical connection **56A** of the substrate **52** by way of a first electrical connection element **16A**.

An (unpackaged) second MEMS device **100B** can be arranged on the underside of the substrate **52** and over the opening **54**. The second MEMS device **100B** can be a photoacoustic detector unit, which can comprise a pressure detector **60** in the form of a MEMS component. The cavities of the second MEMS device **100B** can be filled by a reference gas **64**. The second MEMS device **100B** can be electrically connected to a second electrical connection **56B** of the substrate **52** by way of a second electrical connection element **16B**.

The first MEMS device **100A** or the emitter **58** can be a broadband emitter, which can be designed to emit optical radiation over a wide frequency range. In other words, the radiation emitted by the broadband emitter can comprise not just predetermined frequencies or predetermined frequency bands. The term "optical radiation" used in this description can generally refer to a partial range of the electromagnetic spectrum having wavelengths of between approximately 100 nm and approximately 100 μ m. That is to say that the optical radiation can comprise, in particular, at least one from the following: ultraviolet (UV) radiation having a wavelength of approximately 100 nm to approximately 380 nm, infrared (IR) radiation having a wavelength of approximately 780 nm to approximately 100 μ m, or radiation having a wavelength of approximately 780 nm to approximately 5 μ m, i.e. near-infrared radiation and portions of mid-infrared radiation. The last-mentioned range can comprise, inter alia, the absorption lines/bands of carbon dioxide at 4.26 μ m and of further gas species. Even more specifically, the optical radiation can have a wavelength of approximately 300 nm to approximately 20 μ m (micrometers).

The first MEMS device **100A** can be designed to emit optical pulses having a predetermined repetition frequency

and one or more predetermined wavelengths. In this case, a predetermined wavelength can comprise an absorption band of a gas to be detected or of the reference gas **64**. The repetition frequency of the optical pulses can be within a low-frequency range or within a frequency range of approximately 1 Hz to approximately 10 kHz, in particular of approximately 1 Hz to approximately 1 kHz. Even more specifically, a typical frequency range can be between approximately 1 Hz and approximately 100 Hz, corresponding to a pulse duration range of approximately 0.01 s to approximately 1 s.

A manner of functioning of the photoacoustic sensor **1000** is described below. The further photoacoustic sensors described herein can be operated in a similar manner.

The optical pulses emitted by the first MEMS device **100A** can pass through the interspace formed by the opening **54**, which interspace can be filled with ambient air, for example. During propagation through the opening **54**, the optical pulses can be at least partly absorbed by portions of a gas to be detected if such a gas is present in the opening **54** (i.e. in the ambient air). The absorption can be specific to the gas to be detected, e.g. characteristic rotation or vibration modes of atoms or molecules of the gas to be detected.

The optical pulses can enter the cavity of the MEMS device **100B** through the optically transparent material of the second MEMS device **100B** and impinge there on atoms or molecules of the reference gas **64**. In this case, the reference gas **64** can correspond to the gas to be detected. The optical pulses can at least partly be absorbed by the reference gas **64** and bring about local pressure increases in the reference gas **64**. The pressure increases can be detected by the pressure detector **60** or a movable structure of the pressure detector **60**. The signals detected by the pressure detector **60** can be logically processed by one or more circuits (not illustrated). By way of example, such signal processing can be carried out by an ASIC (Application Specific Integrated Circuit).

If no portions of the gas to be detected are present in the opening **54** or in the ambient air, the optical pulses emitted by the first MEMS device **100A** are only absorbed by the reference gas **64** and the pressure detector **60** will detect a periodic measurement signal with the repetition frequency of the optical pulses and a first amplitude. If, in contrast thereto, portions of the gas to be detected are present in the opening **54**, the optical radiation can additionally be absorbed by these portions. The pressure detector **60** will then output a periodic measurement signal with a second amplitude, which may be smaller than the first amplitude. A presence and/or a concentration of the gas to be detected in the ambient air can be determined on the basis of the magnitudes and profiles of the first and second amplitudes. If the concentration of the gas to be detected exceeds a predetermined threshold value, a signal, in particular a warning signal, can be output by the photoacoustic sensor **1000** or an apparatus connected thereto.

The photoacoustic sensor **1100** in FIG. **11** can be at least partly similar to the photoacoustic sensor **1000** in FIG. **10**. In contrast to FIG. **10**, the photoacoustic emitter unit and the photoacoustic detector unit can be present in the form of packaged MEMS devices **400A** and **400B**, each of which can be similar to the MEMS device **400** in FIG. **4**.

The photoacoustic sensor **1200** in FIG. **12** can comprise a substrate **52**, which can be similar to the substrate **52** in FIG. **10**, for example. Electrical connections and an internal wiring of the substrate **52** are not illustrated in FIG. **12** for the sake of simplicity. A first MEMS device **100A** and a second MEMS device **100B** can be arranged next to one another on the top side of the substrate **52**. The MEMS

devices **100A** and **100B** can each be similar to the MEMS device **100** in FIG. **1** and form a photoacoustic emitter unit and a photoacoustic detector unit. A cover **66** having an optically reflective inner surface **68** can be arranged over the MEMS devices **100A** and **100B**. In this case, the cover **66** can be secured to the substrate **52**. Optical radiation emitted by the first MEMS device **100A** can be reflected once or several times from the inner surface **68** of the cover **66** and can enter the cavity of the second MEMS device **100B**.

The photoacoustic sensor **1300** in FIG. **13** can comprise two MEMS devices **300A** and **300B**, each of which can be similar to the MEMS device **300** in FIG. **3**. The MEMS devices **300A** and **300B** can be arranged next to one another on a substrate **52** in such a way that the openings of the shells **24A** and **24B** can face one another. As a result, optical radiation emitted by the first MEMS device **300A** can pass through an interspace **54** with a gas to be detected possibly being present and can enter the cavity of the second MEMS device **300B**. The MEMS devices **100A** and **100B** arranged in the shells **24A** and **24B** may or may not be encapsulated by an encapsulation material (not illustrated) such as a glob top material, for example.

The photoacoustic sensor **1400** in FIG. **14** can comprise a leadframe **70** having a die pad **72** and one or more connecting conductors (or leads or pins) **74A** and **74B**. A carrier **76** can be arranged on the top side of the die pad **72**, which carrier can be at least partly hollow. A first MEMS device **100A**, which can have the functionality of a photoacoustic emitter unit, can be arranged on the top side of the die pad **72** or on the top side of the carrier **76**. A second MEMS device **100B**, which can have the functionality of a photoacoustic detector unit, can be arranged on the underside of the die pad **72**. The leadframe **70**, the carrier **76** and the MEMS devices **100A**, **100B** can be at least partly encapsulated by an encapsulation material **32**, which can be fabricated from a mold compound, for example. The connecting conductors **74A** and **74B** can at least partly project from the encapsulation material **32** and make the MEMS devices **100A** and **100B** electrically accessible from outside the package by way of electrical connection elements **16A** and **16B**.

The photoacoustic sensor **1400** can comprise a gas channel **78**, which can extend through the encapsulation material **32** and the carrier **76**. In the example in FIG. **14**, the gas channel **78** can have substantially three sections **78A** to **78C**. The first section **78A** can extend in the y-direction substantially from the top side of the encapsulation material **32** as far as the top side of the carrier **76**. The second section **78B** can extend in the x-direction substantially parallel to the die pad **72** through the carrier **76** and can extend between the MEMS devices **100A** and **100B**. The third section **78C** can extend in the y-direction substantially from the underside of the carrier **76** to the underside of the encapsulation material **32**. The sections **78A** to **78C** of the gas channel **78** can merge continuously into one another. In the example in FIG. **14**, the transitions between the sections **78A** to **78C** can have a rounded shape. In further examples, the transitions can have a different shape, for example angular, polygonal, etc. Ambient air can penetrate into the interior of the encapsulation material **32** through the gas channel **78**. An exemplary path that can be taken by the ambient air through the encapsulation material **14** is indicated by arrows in FIG. **14**.

The carrier **76** can have an opening on its top side, such that the underside of the first MEMS device **100A** can be exposed. In a similar manner, the underside of the carrier **76** and the die pad **72** can have openings, such that the top side of the second MEMS device **100B** can be exposed. The

openings of the carrier **76** and of the die pad **72** and the second section **78B** of the gas channel **78** can form an interspace **54** between the MEMS devices **100A** and **100B**. Optical radiation emitted by the first MEMS device **100A** can propagate along an optical path in the y-direction through the interspace **54** to the second MEMS device **100B**. In a similar manner, ambient air can pass through the gas channel **78** into the interspace **54** and thus into the optical path. In accordance with the principles of a photoacoustic sensor described above, the ambient air can be examined for possibly present portions of a gas to be detected.

The photoacoustic sensor **1500** in FIG. **15** can comprise a leadframe **70** having a die pad **72** and connecting conductors **74A** and **74B**. Two MEMS devices **100A** and **100B** can be arranged on the die pad **72**, which MEMS devices can form a photoacoustic emitter unit and a photoacoustic detector unit. The leadframe **70** and the MEMS devices **100A** and **100B** can be embedded in an encapsulation material **32**. The die pad **72** can have openings **80A** and **80B** on its underside. In this case, the first opening **80A** can be arranged below the first MEMS device **100A** in such a way that the underside of the first MEMS device **100A** can be exposed. In a similar manner, the second opening **80B** can be arranged below the second MEMS device **100B** in such a way that the underside of the second MEMS device **100B** can be exposed.

During operation of the photoacoustic sensor **1500**, the first MEMS device **100A** can emit optical radiation through the first opening **80A**. The optical radiation emitted can pass through ambient air, for example, which can be situated below the photoacoustic sensor **1500**, for example. The optical radiation can be reflected at optically reflective structures (not illustrated) in such a way that it can enter the second MEMS device **100B** through the second opening **80B**.

The photoacoustic sensor **1600** in FIG. **16** can be at least partly similar to the photoacoustic sensor **1500** in FIG. **15**. In the example in FIG. **15**, a carrier **76** can be arranged over the die pad **72**. Through the carrier **76**, an optical channel **82** can extend from the underside of the first MEMS device **100A** as far as the underside of the second MEMS device **100B**. Radiation emitted by the first MEMS device **100A** can propagate through the optical channel **82** to the second MEMS device **100B**. In one example, the optical channel **82** can be formed by a hollow channel having optically reflective inner surfaces. The die pad **72** and the carrier **76** can have openings, such that ambient air to be examined can pass into an interspace **54** or into the course of the optical channel **82**.

The photoacoustic sensor **1700** in FIG. **17** can comprise a shell **24**, on the base surface of which a photoacoustic emitter unit and a photoacoustic detector unit in the form of two MEMS devices **100A** and **100B** can be arranged. The MEMS devices **100A** and **100B** can be electrically contacted from outside the shell **24** by way of connecting conductors **26A** and **26B**. During operation of the photoacoustic sensor **1700**, the first MEMS device **100A** can emit optical radiation, for example in the y-direction. The optical radiation emitted can pass through ambient air, for example, which can be situated above the photoacoustic sensor **1700**, for example. The optical radiation can be reflected at optically reflective structures (not illustrated) such that it can enter the second MEMS device **100B**.

The photoacoustic sensor **1800** in FIG. **18** can be at least partly similar to the photoacoustic sensor **1700** in FIG. **17**. In the example in FIG. **18**, a separating structure **84** can be arranged between the MEMS devices **100A** and **100B**.

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Furthermore, a cover **66** having an optically reflective inner surface **68** can be arranged over the MEMS devices **100A** and **100B**.

The photoacoustic sensor **1900** in FIG. **19** can be at least partly similar to the photoacoustic sensor **1700** in FIG. **17**. In contrast to FIG. **17**, the MEMS devices **100A** and **100B** can each be rotated by 90 degrees and be arranged on the base surface of the shell **24**. During operation of the photoacoustic sensor **1900**, the radiation emitted by the photoacoustic emitter can pass substantially in the x-direction, for example.

In the example in FIG. **20**, the MEMS devices **100A** and **100B** can be arranged in a manner stacked one above the other on the base surface of the shell **24**. One or more spacers **86** can be arranged between the MEMS devices **100A** and **100B**, as a result of which an interspace **54** arranged between the MEMS devices **100A** and **100B** can be formed. During operation of the photoacoustic sensor **2000**, ambient air situated in the interspace **54**, for example, can be examined for portions of a gas to be detected. In this case, optical pulses emitted by the first MEMS device **100A** can pass through the interspace **54** in the y-direction.

The MEMS device **2100** in FIG. **21** can comprise a MEMS component **88** and a cover **90** secured to the MEMS component **88**. In this case, the MEMS component **88** and the cover **90** can form a closed cavity **92**. The MEMS component **88** can have an optical opening **94**, which provides an optical access to the cavity **92** and to an optical path extending within the cavity **92**. A movable part or a movable structure **96** of the MEMS component **88** can be arranged outside the course of the optical path.

The MEMS component **88** can be a MEMS chip, for example, which can be fabricated from a semiconductor material such as silicon, for example. The movable part **96** of the MEMS component **88** can be formed from a semiconductor material of the MEMS component **88**. In the example in FIG. **21**, the movable part **96** can form a membrane of a pressure detector. In this case, the MEMS device **2100** can form for example a photoacoustic detector unit of a photoacoustic sensor. Below the movable part **96**, a first depression **98A** can be arranged in a semiconductor material of the MEMS component **88**. An (in particular acoustically tight) cavity can be formed by the first depression **98A**.

The optical opening **94** can be arranged in particular in the underside of the MEMS component **88**. In the example in FIG. **21**, the optical opening **94** can be formed by or comprise a barrier layer **102**. The barrier layer **102** can be fabricated from a semiconductor material of the MEMS component **88**, for example. The optical opening **94** or the barrier layer **102** can be transparent to optical radiation, on the one hand. On the other hand, the optical opening **94** or the barrier layer **102** can be impermeable to a gas arranged in the cavity **92**, in particular a reference gas.

Below the optical opening **94**, a second depression **98B** can be arranged in a semiconductor material of the MEMS component **88**. The depressions **98A** and **98B** may have been produced by identical method steps and have a substantially identical geometric shape. In one example, the first depression **98A** can be fabricated by an etching method in a semiconductor material of the MEMS component **88**. In further process steps, the movable part **96** of the MEMS component **88** can be structured further in order to form a membrane of a pressure detector, for example. The second depression **98B** can be fabricated in parallel with the etching step described. In this case, the further process steps for forming the movable part **96** can be dispensed with, such

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that only the thin barrier layer **102** comprised of a semiconductor material can remain. The barrier layer **102** or the optical opening **94** can thus be regarded as an incompletely produced pressure detector or "half pressure detector". On account of the parallel production described, the optical opening **94** or the barrier layer **102** and the movable part **96** of the MEMS component **88** can lie substantially in an identical plane. In this case, the plane can in particular correspond to the top side of the MEMS component **88** or extend parallel thereto.

The cover **90** can be secured to the top side of the MEMS component **88**, such that the MEMS component **88** and the cover **90** can form the closed cavity **92**. In this case, the cover **90** may have been secured to the MEMS component **88** by means of a wafer bonding technique, in particular. The cover **90** can be fabricated from a glass material or a semiconductor material, for example. The inner surface of the cover **90** can be optically reflective or have optically reflective structures.

In one example, the MEMS device **2100** can be operated in particular as a photoacoustic detector unit. During operation, radiation emitted by an optical emitter (not illustrated) can enter the cavity **92** through the optical opening **94** or the barrier layer **102**. The optical radiation may previously have passed through ambient air, for example, which is intended to be examined for portions of a gas to be detected. The optical radiation can be at least partly absorbed by a reference gas enclosed in the cavity **92**. In this case, the optical radiation can be reflected at the inner surface of the cover **90** in such a way that it can pass at least partly parallel to the top side of the MEMS component **88** after reflection within the cavity **92**. On account of such a parallel course, a lengthened absorption path can be provided in comparison with conventional photoacoustic detector units. The optical radiation can be at least partly absorbed by the reference gas and bring about local pressure increases in the reference gas. The pressure increases in the reference gas can be detected by the movable part **96**. The detected signals can be logically processed for example by an ASIC (not illustrated).

In contrast to conventional photoacoustic detector units, in the example in FIG. **21** the movable part **96** can be arranged such that it lies outside a course of the optical radiation, i.e. outside the optical path. Consequently, since the optical radiation does not impinge on a surface of the movable part **96**, corruption of the signals detected by the movable part **96** can be avoided or at least reduced.

FIGS. **22** and **23** show photoacoustic sensors **2200** and **2300**, each of which can comprise a photoacoustic emitter unit and a photoacoustic detector unit. The emitter unit and detector unit can be realized in each case by a MEMS device which can be similar to the MEMS device **2100** in FIG. **21**. In further examples, the MEMS devices **2100** can also be combined to form a photoacoustic sensor in accordance with any of FIGS. **10** to **20**.

The photoacoustic sensor **2200** in FIG. **22** can be at least partly similar to the photoacoustic sensor **2000** in FIG. **20** and be operated in a similar manner. Two MEMS devices **2100A** and **2100B** stacked one above the other can be arranged on the base surface of a shell **24**, which MEMS devices can be spaced apart from one another by one or more spacers **86**. The lower MEMS device **2100A** can have the function of a photoacoustic emitter unit. In this case, a movable part **96** of the MEMS device **2100A** can form a heating memory, for example, which can be designed to emit optical radiation. The upper MEMS device **2100B** can have the function of a photoacoustic detector unit and can be similar to the MEMS device **2100** in FIG. **21**.

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The photoacoustic sensor **2300** in FIG. **23** can comprise a housing **104**, which can be embodied in a U-shaped fashion, for example. In the example in FIG. **23**, the housing **104** can have two substantially vertical sections **106A** and **106B**, which can be connected by a substantially horizontal section **106C**. The vertical housing sections **106A** and **106B** can have cavities **108A** and **108B**. A first MEMS device **2100A**, which can be embodied as a photoacoustic emitter unit, can be arranged in the first cavity **108A**. A second MEMS device **2100B**, which can be embodied as a photoacoustic detector unit, can be arranged in the second cavity **108B**.

The vertical housing sections **106A** and **106B** can have openings **110A** and **110B**, which can be arranged at an identical height and can face one another. The MEMS devices **2100A** and **2100B** can be arranged in an analogous manner such that their main surfaces can face one another. During operation of the photoacoustic sensor **2300**, optical radiation emitted by the first MEMS device **2100A** can pass through the opening **110A** and an interspace **54** situated between the vertical housing sections **106A** and **106B**. In the interspace **54**, the optical radiation can be absorbed by a gas to be detected possibly being present there and can enter the second MEMS device **2100B** through the second opening **108B**.

EXAMPLES

Photoacoustic sensors and MEMS devices are explained below on the basis of examples.

Example 1 is a photoacoustic sensor, comprising:

a first MEMS device, comprising: a first MEMS component comprising an optical emitter, and a first optically transparent cover wafer-bonded to the first MEMS component, wherein the first MEMS component and the first optically transparent cover form a first closed cavity; and a second MEMS device, comprising: a second MEMS component comprising a pressure detector, and a second optically transparent cover wafer-bonded to the second MEMS component, wherein the second MEMS component and the second optically transparent cover form a second closed cavity

Example 2 is a photoacoustic sensor according to example 1, wherein the first MEMS device further comprises: a third cover wafer-bonded to the first MEMS component, wherein the first MEMS component and the third cover form a third closed cavity, wherein the first cavity and the third cavity are arranged on opposite sides of a movable structure of the first MEMS component.

Example 3 is a photoacoustic sensor according to example 1, wherein the first MEMS device further comprises: a carrier, wherein the first cover is wafer-bonded to the carrier, wherein the first cavity is arranged between the first cover and a movable structure of the first MEMS component, and wherein a third cavity is arranged between the carrier and the movable structure.

Example 4 is a photoacoustic sensor according to any of the preceding examples, furthermore comprising: a substrate, wherein the first MEMS device and the second MEMS device are arranged next to one another on a same surface of the substrate.

Example 5 is a photoacoustic sensor according to example 4, furthermore comprising: a cover arranged above the MEMS devices arranged next to one another, said cover having an optically reflective inner surface.

Example 6 is a photoacoustic sensor according to any of examples 1 to 3, furthermore comprising: a substrate,

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wherein the first MEMS device and the second MEMS device are arranged on opposite surfaces of the substrate and over an opening formed in the substrate.

Example 7 is a photoacoustic sensor according to any of examples 1 to 3, furthermore comprising: an encapsulation material, wherein the first MEMS device and the second MEMS device are encapsulated by the encapsulation material.

Example 8 is a photoacoustic sensor according to example 7, furthermore comprising: a gas channel extending within the encapsulation material between the first MEMS device and the second MEMS device.

Example 9 is a photoacoustic sensor according to example 7 or 8, furthermore comprising: an optical path extending within the encapsulation material from the first MEMS device to the second MEMS device.

Example 10 is a photoacoustic sensor according to example 9, furthermore comprising: a leadframe, wherein the first MEMS device and the second MEMS device are arranged on opposite surfaces of the leadframe, wherein the gas channel extends at least partly parallel to the surfaces of the leadframe, and wherein the optical path extends at least partly perpendicularly to the surfaces of the leadframe.

Example 11 is a photoacoustic sensor according to example 9, furthermore comprising: leadframe, wherein the first MEMS device and the second MEMS device are arranged on a same surface of the leadframe, wherein the optical path extends at least partly parallel to the surface of the leadframe.

Example 12 is a photoacoustic sensor according to any of examples 1 to 3, furthermore comprising: a shell, wherein the first MEMS device and the second MEMS device are arranged next to one another on a base of the shell.

Example 13 is a photoacoustic sensor according to any of examples 1 to 3, furthermore comprising: a shell, wherein the first MEMS device and the second MEMS device are arranged in a manner stacked one above the other on a base of the shell and a spacer is arranged between the MEMS devices stacked one above the other.

Example 14 is a MEMS device, comprising: a MEMS component; a cover secured to the MEMS component, wherein the MEMS component and the cover form a closed cavity; and an optical opening, which provides an optical access to the cavity and to an optical path extending within the cavity, wherein a movable part of the MEMS component is arranged outside the course of the optical path.

Example 15 is a MEMS device according to example 14, wherein the optical opening and the movable part of the MEMS component lie substantially in a same plane.

Example 16 is a MEMS device according to example 14 or 15, wherein the optical opening comprises a barrier layer formed by a semiconductor material of the MEMS component.

Example 17 is a MEMS device according to any of examples 14 to 16, wherein the optical opening is impermeable to a gas arranged in the cavity.

Example 18 is a MEMS device according to any of examples 14 to 17, wherein the cover is produced from a glass material and is secured to a semiconductor material of the MEMS component.

Example 19 is a MEMS device according to example 18, wherein the glass material of the cover is wafer-bonded to the semiconductor material of the MEMS component.

Example 20 is a MEMS device according to any of examples 14 to 19, wherein: the cover is arranged over a first surface of the MEMS component, and the optical opening is

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formed in a second surface of the MEMS component situated opposite the first surface.

Example 21 is a MEMS device according to example 20, wherein the optical path extends within the cavity at least partly parallel to the first surface of the MEMS component.

Example 22 is a MEMS device according to any of examples 14 to 21, furthermore comprising: a first depression formed in a semiconductor material of the MEMS component, said first depression being arranged below the movable part of the MEMS component; a second depression formed in a semiconductor material of the MEMS component, said second depression being arranged below the optical opening, wherein the first depression and the second depression have a substantially identical geometric shape.

Example 23 is a MEMS device according to any of examples 14 to 22, wherein the MEMS component comprises a pressure detector.

Although specific embodiments have been illustrated and described herein, it is obvious to the person of average skill in the art that a multiplicity of alternative and/or equivalent implementations can replace the specific embodiments shown and described, without departing from the scope of the present disclosure. This application is intended to cover all adaptations or variations of the specific embodiments discussed herein. Therefore, the intention is for this disclosure to be restricted only by the claims and the equivalents thereof.

What is claimed is:

1. A MEMS device, comprising:
 - a MEMS component comprising:
 - a first surface;
 - a second surface situated opposite to the first surface;
 - a barrier layer arranged at the first surface, the barrier layer being transparent to optical radiation and impermeable to a reference gas; and
 - a movable part arranged at the first surface;
 - a cover secured to the first surface of the MEMS component, wherein the MEMS component and the cover form a closed cavity over the first surface of the MEMS component, and wherein the barrier layer and the movable part are arranged at a boundary of the closed cavity and are exposed to the reference gas, which is contained in the closed cavity; and
 - an optical opening that extends from the second surface of the MEMS component to the barrier layer, wherein the optical opening provides an optical access to the closed cavity and to an optical path extending within the closed cavity, wherein the movable part of the MEMS component is arranged outside a course of the optical path.
2. The MEMS device as claimed in claim 1, wherein the optical opening and the movable part of the MEMS component lie substantially in an identical plane.
3. The MEMS device as claimed in claim 1, wherein the cover is produced from a glass material and is secured to a semiconductor material of the MEMS component.
4. The MEMS device as claimed in claim 3, wherein the glass material of the cover is wafer-bonded to the semiconductor material of the MEMS component.
5. The MEMS device as claimed in claim 1, wherein the optical path extends within the closed cavity at least partly parallel to the first surface of the MEMS component.
6. The MEMS device as claimed in claim 1, further comprising:

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a first depression formed in a semiconductor material of the MEMS component, the first depression being arranged below the movable part of the MEMS component, and the movable part being arranged between the closed cavity and the first depression; and

a second depression formed in the semiconductor material of the MEMS component, the second depression being arranged below the barrier layer, the second depression defining a portion of the optical opening.

7. The MEMS device as claimed in claim 1, wherein the MEMS component comprises a pressure detector formed, in part, by the movable part.

8. The MEMS device as claimed in claim 1, wherein the cover is a transparent cover.

9. The MEMS device as claimed in claim 1, wherein the cover is wafer-bonded to the MEMS component.

10. The MEMS device as claimed in claim 1, wherein the cover is a first cover,

wherein the closed cavity is a first closed cavity, and wherein the MEMS device further comprises:

a second cover that forms a second closed cavity with the MEMS component.

11. The MEMS device as claimed in claim 10, wherein the second cover is optically transparent.

12. A photoacoustic detector device, comprising:

a MEMS component comprising:

a first surface;

a second surface situated opposite to the first surface;

a barrier layer, the barrier layer being transparent to optical radiation and impermeable to a gas; and

a movable membrane;

a cover,

wherein the MEMS component and the cover form a cavity over the first surface of the MEMS component,

wherein the barrier layer and the movable membrane are coupled to the cavity; and

an opening that is configured to provide, via the barrier layer, an optical access to the cavity and to an optical path within the cavity,

wherein the movable membrane of the MEMS component is arranged outside the optical path.

13. The MEMS device as claimed in claim 1, wherein the MEMS component is a MEMS chip fabricated from a semiconductor material, and

wherein the barrier layer and the movable part are formed from the semiconductor material.

14. The MEMS device as claimed in claim 6, wherein the first depression and the second depression have a substantially identical geometric shape.

15. The MEMS device as claimed in claim 1, wherein an inner surface of the cover, arranged within the closed cavity, is optically reflective or has optically reflective structures.

16. The MEMS device as claimed in claim 1, wherein the movable part of the MEMS component is arranged outside the course of the optical path such that the optical radiation does not impinge on a surface of the movable part.

17. The photoacoustic detector device as claimed in claim 12, wherein the opening extends from the second surface of the MEMS component to the barrier layer, and

wherein the opening is configured to provide the optical access, to the cavity, for the optical radiation such that the optical radiation propagates along the optical path.

18. The photoacoustic detector device as claimed in claim 12, wherein the opening and the movable membrane of the MEMS component lie substantially in an identical plane.

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19. The photoacoustic detector device as claimed in claim 12, further comprising:
- a first depression formed in a semiconductor material of the MEMS component, the first depression being arranged below the movable membrane of the MEMS component, and the movable membrane being arranged between the cavity and the first depression; and
 - a second depression formed in the semiconductor material of the MEMS component, the second depression being arranged below the barrier layer, the second depression defining a portion of the opening.
20. The photoacoustic detector device as claimed in claim 12, wherein the cavity is a closed cavity that contains a reference gas as the gas.

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